



Modelling Financial Markets

Seminar 8: Backtesting and Evaluating Risk Forecasts — Hands-on



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Acronyms used in this material

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CLT: Central Limit Theorem
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ POF: Proportion Of Failures (test)

Outline

Introduction

Key Formulas

Part A: Derivations

Part B: Computation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the asymptotics, size and power of VaR backtests and the consistency of scoring functions
- ▣ Backtest ES with residuals and the Acerbi–Szekely Z_2 by simulation
- ▣ Rank (VaR, ES) forecasts with the FZ0 loss, Diebold–Mariano tests and the MCS
- ▣ Build a conformal VaR and check its coverage in calm and crisis periods

- ▣ Background: Lecture 8; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** ten derivations (A1–A10)
- ▣ **Part B:** nine computational exercises with inference (B1–B9)
- ▣ **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Report every test with its p -value and say how it was obtained (asymptotic, exact, bootstrap, simulation)
- A model that passes a test is not validated: ask how much power the test had

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (POF asymptotics and power), A3 (Markov-chain likelihood), A5 (traffic light), A6 (consistency of pinball and FZ0), A7 ($E[Z_2] = 0$), A8 (conformal quantile, ACI)
- Part B: B1 (VaR tests, DQ and size, S&P 500), B3 (ES tests with simulation), B5 (Diebold–Mariano with HAC, Giacomini–White), B7 (conformal VaR in crises)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A9 (model: A6), A10 (model: A7)
- Part B: B2 (model: A5, B1), B4 (model: B3), B6 (model: B5), B8 (model: B7), B9 (model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** `chapter8_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: S&P 500, BET, Bitcoin (daily closes); EUR/RON: BNR reference rate; up to 18 September 2026
- Forecasts: the six models of Lecture 8 (HS, Normal, Student-t, GARCH-t, FHS, GARCH-EVT), rolling window of 1000 days, parameters re-estimated every 20 days
- Losses: $L_t = -r_t$ with $r_t = \ln P_t - \ln P_{t-1}$; breach: $I_t = 1\{L_t > \text{VaR}_t\}$

Key Formulas (1/2): VaR Backtests

- Kupiec [Kupiec \(1995\)](#): $LR_{uc} = -2 [(T - x) \ln(1 - p) + x \ln p - (T - x) \ln(1 - \hat{\pi}) - x \ln \hat{\pi}]$, $\hat{\pi} = x/T$, χ_1^2
- Christoffersen [Christoffersen \(1998\)](#): $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$, $\hat{\pi} = \frac{n_{01} + n_{11}}{n}$
- $LR_{ind} = -2 [(n_{00} + n_{10}) \ln(1 - \hat{\pi}) + (n_{01} + n_{11}) \ln \hat{\pi} - n_{00} \ln(1 - \hat{\pi}_{01}) - n_{01} \ln \hat{\pi}_{01} - n_{10} \ln(1 - \hat{\pi}_{11}) - n_{11} \ln \hat{\pi}_{11}]$
- $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi_2^2$; durations [Christoffersen & Pelletier \(2004\)](#): Weibull shape $b = 1$ under H_0
- DQ [Engle & Manganelli \(2004\)](#): $\text{Hit}' X (X' X)^{-1} X' \text{Hit} / (\alpha(1 - \alpha)) \sim \chi_k^2$, $\text{Hit}_t = I_t - \alpha$, $X_t \in \mathcal{F}_{t-1}$
- Traffic light [BCBS \(1996\)](#): $X \sim \text{Binomial}(250, 0.01)$; yellow from $P(X \leq x) \geq 95\%$, red from $P(X \leq x) \geq 99.99\%$

Key Formulas (2/2): ES, Scores and Conformal VaR

- McNeil–Frey [McNeil & Frey \(2000\)](#): $e_t = (L_t - \text{ES}_t)/\sigma_t$ on breach days; $H_0: E[e_t] = 0$, $H_1: E[e_t] > 0$
- Acerbi–Szekely [Acerbi & Szekely \(2014\)](#): $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t \hat{L}_t}{\text{ES}_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t \hat{L}_t}{\text{ES}_t}$; fixed thresholds -0.70 and -1.8 for $T = 250$
- FZ0 [Patton, Ziegel & Chen \(2019\)](#): $\frac{\mathbf{1}_{\{L > v\}}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$, with VaR v and ES e (positive losses)
- Diebold–Mariano [Diebold & Mariano \(1995\)](#): $\bar{d}/\sqrt{\widehat{\text{LRV}}/T}$ with Newey–West weights [Newey & West \(1987\)](#); MCS [Hansen, Lunde & Nason \(2011\)](#); Giacomini–White [Giacomini & White \(2006\)](#): $T \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \sim \chi_q^2$, $Z_t = h_{t-1} d_t$
- Conformal VaR: $\hat{\sigma}_t s_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$; ACI [Gibbs & Candès \(2021\)](#): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$

A1: Asymptotics and Power of the POF Test [Solved]

Task

- **Context:** no data; a VaR 1% forecast is evaluated on T days, and the hits $I_t = 1\{L_t > \text{VaR}_t\}$ are i.i.d. Bernoulli(π).
 - ▶ Notation: $x = \sum_t I_t$, $\hat{\pi} = x/T$, log-likelihood $\ell(\pi) = x \ln \pi + (T - x) \ln(1 - \pi)$, Kupiec statistic $\text{LR}_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$ Kupiec (1995).
- (a) Expand $\ell(\alpha)$ to second order around $\hat{\pi}$ and show that, under $H_0: \pi = \alpha$, LR_{uc} converges in distribution to χ_1^2 .
- (b) Let the true probability be the local alternative $\pi_T = \alpha + h/\sqrt{T}$; derive the limit of LR_{uc} and give its noncentrality parameter λ .
- (c) For $T = 250$, $\alpha = 1\%$ and $\pi = 2\%$, compute the local power $P(\chi_1^2(\lambda) > 3.84)$, the exact Binomial power of the 5% test and its exact size.
- **Report** λ , the two powers and the size, and **answer:** can one year of VaR 1% data detect a model that doubles the breach rate?

A1: Asymptotics and Power of the POF Test: Solution (1/2) [Solved]

Solution

- (a) Step 1: $\ell'(\hat{\pi}) = x/\hat{\pi} - (T - x)/(1 - \hat{\pi}) = 0$ at the MLE, so the first-order term of the expansion vanishes.
- Step 2: $\ell(\alpha) = \ell(\hat{\pi}) + \frac{1}{2}\ell''(\bar{\pi})(\alpha - \hat{\pi})^2$ with $\bar{\pi}$ between α and $\hat{\pi}$, and $-\ell''(\bar{\pi})/T = \frac{\hat{\pi}}{\bar{\pi}^2} + \frac{1-\hat{\pi}}{(1-\bar{\pi})^2} \xrightarrow{P} \frac{1}{\alpha(1-\alpha)}$.
- Step 3: therefore $\text{LR}_{uc} = \frac{T(\hat{\pi} - \alpha)^2}{\alpha(1 - \alpha)} + o_p(1) = Z_T^2 + o_p(1)$ with $Z_T = \frac{\sqrt{T}(\hat{\pi} - \alpha)}{\sqrt{\alpha(1 - \alpha)}}$.
- Step 4: by the CLT for i.i.d. Bernoulli variables $Z_T \xrightarrow{d} N(0, 1)$, hence $\text{LR}_{uc} \xrightarrow{d} \chi_1^2$; the 5% critical value is 3.84.
- (b) Under π_T : $\sqrt{T}(\hat{\pi} - \alpha) = \sqrt{T}(\hat{\pi} - \pi_T) + h$, so $Z_T \xrightarrow{d} N(h/\sqrt{\alpha(1 - \alpha)}, 1)$ and $\text{LR}_{uc} \xrightarrow{d} \chi_1^2(\lambda)$ with

$$\lambda = \frac{h^2}{\alpha(1 - \alpha)} = \frac{T(\pi - \alpha)^2}{\alpha(1 - \alpha)}.$$

A1: Asymptotics and Power of the POF Test: Solution (2/2) [Solved]

Solution

- (c) Noncentrality: $\lambda = \frac{250(0.02 - 0.01)^2}{0.01 \cdot 0.99} = \frac{0.025}{0.0099} = 2.53$; local power $P(\chi_1^2(2.53) > 3.84) = 36\%$.
- Exact test: $LR_{uc}(x) \leq 3.84$ exactly for $x \in \{1, \dots, 6\}$, so the acceptance region is $1 \leq x \leq 6$.
- Exact power: $1 - P(1 \leq X \leq 6 \mid X \sim \text{Bin}(250, 0.02)) = 1 - 75.7\% = 24\%$.
- Exact size: $P(X = 0) + P(X \geq 7)$ under $\text{Bin}(250, 0.01) = 8.11\% + 1.37\% = 9.5\%$, almost twice the nominal 5%.
- Interpretation: with $\alpha T = 2.5$ expected breaches the lattice of x dominates; a model that doubles the breach rate passes one year of testing about three times out of four.

A2: Power of the POF Test at the FRTB Desk Level [Proposed]

Solution discussed in class

Task

- **Context:** no data; a trading desk reports VaR 2.5% ($\alpha = 0.025$, the FRTB desk level) for one year, $T = 250$; hits are i.i.d. Bernoulli(π).
- (a) Compute the noncentrality λ and the local power of the 5% Kupiec test against $\pi = 5\%$; then find the exact acceptance region of x and compute the exact Binomial power and size.
- (b) Find λ^* with $P(\chi_1^2(\lambda^*) > 3.84) = 0.80$ and solve $T(\pi - \alpha)^2 / (\alpha(1 - \alpha)) = \lambda^*$ for the number of days T ; then find the first T at which the exact power reaches 80%.
- (c) Compute $P(X > 30)$ for $X \sim \text{Bin}(250, 0.05)$, the probability that a desk with a doubled breach rate exceeds the FRTB limit of 30 exceptions.
- **Report** λ , the powers, the size, the two values of T and $P(X > 30)$; **answer:** what does the FRTB desk limit actually screen for?
- Model: A1

A3: Christoffersen Test and the Markov-Chain Likelihood [Solved]

Task

- ▣ **Context:** BET index (data/market/BET.csv), daily VaR 1% of historical simulation (window 1000 days), 2007–2010, 1005 days.
 - ▶ Transition counts of the hit sequence: $n_{00} = 958$, $n_{01} = 22$, $n_{10} = 22$, $n_{11} = 2$ (n_{ij} : days in state j after a day in state i); $LR_{uc} = 14.11$ is given.
- ▣ (a) Compute the transition probabilities $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$ and the pooled $\hat{\pi} = (n_{01} + n_{11})/n$.
- ▣ (b) Compute the independence statistic LR_{ind} Christoffersen (1998), its χ^2_1 p -value and the decision at 5%.
- ▣ (c) Compute $LR_{cc} = LR_{uc} + LR_{ind}$, its χ^2_2 p -value and the decision at 5%.
- ▣ (d) Write the likelihood of the first-order Markov chain conditional on I_1 , derive $\hat{\pi}_{01}$ and $\hat{\pi}_{11}$ as MLEs, and explain why LR_{cc} has 2 degrees of freedom.
- ▣ **Report** the three probabilities, both statistics with p -values, and **answer**: does HS fail because of too many breaches or because breaches cluster?

A3: Christoffersen Test and the Markov-Chain Likelihood: Solution (1/2) [Solved]

Solution

- (a) $\hat{\pi}_{01} = 22/(958 + 22) = 0.0224$; $\hat{\pi}_{11} = 2/(22 + 2) = 0.0833$; $\hat{\pi} = (22 + 2)/1004 = 0.0239$.
- After a breach the breach probability is almost four times higher (0.0833 versus 0.0224).
- (b) $LR_{ind} = -2[\ln L(\hat{\pi}, \hat{\pi}) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})] = 2.32$, $p = 0.128 > 0.05$: independence is not rejected, because $\hat{\pi}_{11}$ rests on only 2 consecutive breaches.
- (c) $LR_{cc} = 14.11 + 2.32 = 16.43$, $p = 0.0003 < 0.05$: conditional coverage is rejected, driven by the coverage part.
- Interpretation: HS fails mainly through too many breaches ($\hat{\pi} \approx 2.4\%$ instead of 1%); the clustering is visible but the test lacks the data to confirm it.

A3: Christoffersen Test and the Markov-Chain Likelihood: Solution (2/2) [Solved]

Solution

- (d) Step 1: conditional on I_1 , $\ln L(\pi_{01}, \pi_{11}) = n_{00} \ln(1 - \pi_{01}) + n_{01} \ln \pi_{01} + n_{10} \ln(1 - \pi_{11}) + n_{11} \ln \pi_{11}$.
- Step 2: the log-likelihood separates into a part in π_{01} and a part in π_{11} ; setting $\partial/\partial\pi_{i1} = -n_{i0}/(1 - \pi_{i1}) + n_{i1}/\pi_{i1} = 0$ gives $\hat{\pi}_{i1} = n_{i1}/(n_{i0} + n_{i1})$, two Bernoulli MLEs.
- Step 3: $LR_{cc} = -2[\ln L(\alpha, \alpha) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})]$ fixes both free parameters, $\pi_{01} = \alpha$ and $\pi_{11} = \alpha$: two restrictions, hence χ^2_2 .
- Step 4: adding and subtracting $\ln L(\hat{\pi}, \hat{\pi})$ splits it exactly into $LR_{uc} + LR_{ind}$ when both are computed on the same $T - 1$ transitions; if LR_{uc} uses all T days, the first observation enters only one term and the identity holds only asymptotically.

A4: The DQ Statistic as a Wald Test [Proposed]

Solution discussed in class

Task

- **Context:** no data; $\text{Hit}_t = I_t - \alpha$ and k regressors X_t known at $t - 1$ (constant, lagged hits, the VaR level); linear probability model $\text{Hit}_t = X_t\beta + u_t$ [Engle & Manganelli \(2004\)](#).
 - ▶ Null hypothesis: correct conditional coverage, $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$.
- (a) Show that under H_0 $E[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$ and $\text{Var}(\text{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$.
- (b) Using a martingale-difference CLT for $T^{-1/2}X'\text{Hit}$, derive $\text{DQ} = \text{Hit}'X(X'X)^{-1}X'\text{Hit}/(\alpha(1 - \alpha)) \xrightarrow{d} \chi_k^2$.
- (c) Take $X_t = 1$ only: show that DQ becomes $T(\hat{\pi} - \alpha)^2/(\alpha(1 - \alpha))$, compute it for $T = 250$, $x = 7$, $\alpha = 1\%$ with its χ_1^2 p -value, and compare with $\text{LR}_{uc} = 5.50$ ($p = 0.019$).
- **Report** the derivation and the two statistics; **answer:** why do two asymptotically equivalent tests disagree here?
- Model: A3

A5: The Traffic Light from the Binomial Distribution [Solved]

Task

- ▣ **Context:** no data; the Basel traffic light [BCBS \(1996\)](#) counts the exceptions X of a VaR 1% over the last 250 days; for a correct model $X \sim \text{Binomial}(250, 0.01)$.
- ▣ (a) Compute $P(X \leq x)$ for $x = 4, 5, 9, 10$ and derive the zones: green while $P(X \leq x) < 95\%$, red from $P(X \leq x) \geq 99.99\%$, yellow in between.
- ▣ (b) A bank reports 6 exceptions: give the zone, the 1996 plus factor, the FRTB capital multiplier and the probability $P(X \geq 6)$ for a correct model.
- ▣ (c) Compute the probability that a model with a true breach probability of 3% still lands in the green zone, $P(X \leq 4 \mid p = 0.03)$.
- ▣ **Report** the four probabilities, the zones and the answers to (b) and (c); **answer:** which error does the traffic light control, and which does it leave large?

A5: The Traffic Light from the Binomial Distribution: Solution [Solved]

Solution

- ▣ (a) $P(X \leq 4) = 89.22\%$, $P(X \leq 5) = 95.88\%$, $P(X \leq 9) = 99.97\%$, $P(X \leq 10) = 99.99\%$.
- ▣ The 95% threshold is first crossed at $x = 5$ and 99.99% at $x = 10$: green 0–4, yellow 5–9, red 10 or more.
- ▣ (b) 6 exceptions: yellow zone; 1996 plus factor 0.50 (multiplier 3.50); FRTB multiplier 1.76;
 $P(X \geq 6) = 1 - P(X \leq 5) = 4.12\%$: a correct model rarely produces this many.
- ▣ (c) $P(X \leq 4 \mid p = 0.03) = 12.8\%$: a model that triples the breach rate stays green about one time in eight.
- ▣ Interpretation: the zones keep the type I error small (about 5% of correct models leave green), but the type II error against bad models remains large with 250 days.

A6: Consistency of the Pinball and FZ0 Losses [Solved]

Task

- ▣ **Context:** a loss L with continuous cdf F and density f ; tail probability $\alpha \in (0, 1)$; forecasts v (VaR) and e (ES), with $e \geq v > 0$.
- ▣ (a) Differentiate the expected pinball loss $E[(1\{L > v\} - \alpha)(L - v)]$ in v and show that its minimiser is VaR_α .
- ▣ (b) For $\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$ [Patton, Ziegel & Chen \(2019\)](#), compute $\partial_v E[\text{FZ0}]$ and $\partial_e E[\text{FZ0}]$, show that both vanish at $(\text{VaR}_\alpha, \text{ES}_\alpha)$ and write the two identification functions.
- ▣ (c) Check numerically: simulate 10^6 Student- t_4 losses with unit variance, minimise the average pinball loss and the average FZ0 loss at $\alpha = 2.5\%$, and compare with the true VaR and ES.
- ▣ **Report** the derivatives, the identification functions and the numerical minimisers; **answer:** why does consistency make ranking by average score meaningful?

A6: Consistency of the Pinball and FZ0 Losses: Solution [Solved]

Solution

- (a) $\frac{d}{dv} E[(1\{L > v\} - \alpha)(L - v)] = -P(L > v) + \alpha = F(v) - (1 - \alpha)$; setting it to zero gives $F(v) = 1 - \alpha$, i.e. $v = \text{VaR}_\alpha$; the second derivative $f(v) > 0$ confirms a minimum.
- (b) $\partial_v E[\text{FZ0}] = \frac{1}{e} \left(1 - \frac{P(L > v)}{\alpha} \right)$, zero exactly when $P(L > v) = \alpha$, i.e. $v = \text{VaR}_\alpha$.
- $\partial_e E[\text{FZ0}] = \frac{1}{e^2} \left(e - v - \frac{E[(L - v)1\{L > v\}]}{\alpha} \right)$; at $v = \text{VaR}_\alpha$, $E[(L - v)1\{L > v\}]/\alpha = E[L - v \mid L > v]$, so $e = E[L \mid L > v] = \text{ES}_\alpha$.
- Identification functions: $V_1(v, L) = \alpha - 1\{L > v\}$ and $V_2(v, e, L) = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$; their expectations vanish only at the true pair (Fissler & Ziegel (2016)).
- (c) Numerical minimisers: pinball $v = 1.960$; FZ0 $(v, e) = (1.960, 2.812)$; truth $(1.963, 2.824)$: equal up to simulation error.
- Interpretation: a forecaster who knows F minimises the expected score by reporting the truth, so a lower average score is evidence of a better forecast.

A7: Why $E[Z_2] = 0$, and Z_1, Z_2 on One Year [Solved]

Task

- **Context:** ES 2.5% forecasts ($\alpha = 0.025$) evaluated with the Acerbi–Szekely statistics Acerbi & Szekely (2014);
 $l_t = 1\{L_t > \text{VaR}_t\}$, $N_T = \sum_t l_t$.
 ► $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t l_t}{\text{ES}_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t l_t}{\text{ES}_t}$
- (a) Assuming a continuous and correct forecast distribution F_t , prove $E[L_t l_t \mid \mathcal{F}_{t-1}] = \alpha \text{ES}_t$ and deduce $E[Z_2] = 0$; explain on what Z_1 conditions.
- (b) One year, $T = 250$, 8 breach days with losses (%) 3.9, 2.8, 4.6, 3.1, 5.2, 2.9, 3.6, 7.4 and ES forecasts (%) 3.4, 3.3, 3.6, 3.2, 3.5, 3.3, 3.4, 3.8: compute the ratios L_t/ES_t , Z_1 and Z_2 , and classify Z_2 with the thresholds -0.70 (yellow) and -1.8 (red).
- (c) Recompute Z_2 after adding one more breach with $L = 7\%$ and $\text{ES} = 3.5\%$.
- **Report** the proof, Z_1 , Z_2 and the zone; **answer:** which days drive Z_2 ?

A7: Why $E[Z_2] = 0$, and Z_1, Z_2 on One Year: Solution [Solved]

Solution

- (a) $E[L_t 1\{L_t > \text{VaR}_t\} \mid \mathcal{F}_{t-1}] = P(L_t > \text{VaR}_t \mid \mathcal{F}_{t-1}) E[L_t \mid L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = \alpha \text{ES}_t$.
- ES_t is known at $t - 1$, so $E[L_t / \text{ES}_t \mid \mathcal{F}_{t-1}] = \alpha$ for every t and $E[Z_2] = 1 - T\alpha / (T\alpha) = 0$: the denominator $T\alpha$ is a constant.
- Z_1 divides by the random N_T : it has mean zero only conditionally on $N_T \geq 1$, is undefined when $N_T = 0$, and ignores how many breaches occurred.
- (b) Ratios: 1.147, 0.848, 1.278, 0.969, 1.486, 0.879, 1.059, 1.947; their sum is 9.613.
- $Z_1 = 1 - 9.613/8 = -0.202$: tail losses are on average 20% above the forecast ES ;
 $Z_2 = 1 - 9.613/6.25 = -0.538 > -0.70$: green zone.
- (c) With one more ratio $7/3.5 = 2$: $Z_2 = -0.86 < -0.70$, yellow. Interpretation: Z_2 reacts to both the number and the size of breaches, and a single large loss can change the zone.

A8: Conformal Quantile and One ACI Step [Solved]

Task

- ▣ **Context:** $n = 19$ calibration scores $s = L/\hat{\sigma}$ (sorted): 0.19, 0.35, 0.52, 0.61, 0.73, 0.88, 0.94, 1.05, 1.12, 1.28, 1.47, 1.66, 1.83, 1.95, 2.02, 2.28, 2.41, 2.75, 3.10; tomorrow's volatility forecast is $\hat{\sigma} = 1.4\%$.
- ▣ (a) Compute the split conformal VaR 10%: the order statistic $s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$, times $\hat{\sigma}$.
- ▣ (b) Apply one ACI step [Gibbs & Candès \(2021\)](#) with $\gamma = 0.02$ and $\alpha_1 = 0.10$: compute $\alpha_2 = \alpha_1 + \gamma(\alpha - \text{err}_1)$ after a breach ($\text{err}_1 = 1$) and after no breach ($\text{err}_1 = 0$).
- ▣ (c) Recompute the order index and the VaR for both values of α_2 .
- ▣ **Report** k , the three VaR values and **answer**: how smoothly does ACI move the VaR?

A8: Conformal Quantile and One ACI Step: Solution [Solved]

Solution

- (a) $k = \lceil 20 \cdot 0.9 \rceil = 18$; $s_{(18)} = 2.75$; $\text{VaR} = 1.4 \cdot 2.75 = 3.85\%$.
- (b) After a breach: $\alpha_2 = 0.10 + 0.02(0.10 - 1) = 0.082$; after no breach: $\alpha_2 = 0.10 + 0.02 \cdot 0.10 = 0.102$.
- (c) After a breach: $k = \lceil 20(1 - 0.082) \rceil = \lceil 18.36 \rceil = 19$, $\text{VaR} = 1.4 \cdot 3.10 = 4.34\%$, a higher VaR.
- After no breach: $k = \lceil 17.96 \rceil = 18$, the same index, so the VaR does not move.
- Interpretation: with small n ACI moves in jumps of one order statistic; small changes of α_t are invisible until they cross a grid point $j/(n+1)$.

A9: ES Alone Is Not Elicitable [Proposed]

Solution discussed in class

Task

- **Context:** a functional $T(F)$ is elicitable if $T(F) = \arg \min_x E_F[S(x, L)]$ for some scoring function S [Gneiting \(2011\)](#); here $T = \text{ES}_\alpha$ with $\alpha < 1/2$.
- (a) Prove that if T is elicitable, every level set $\{F : T(F) = t\}$ is convex, i.e. closed under mixtures $F_\lambda = (1 - \lambda)F_0 + \lambda F_1$.
- (b) Let $F_0 = \delta_1$ (loss 1 with certainty) and F_1 with $P(L = 2) = \alpha/2$, $P(L = 0) = 1 - \alpha/2$; compute ES_α of F_0 , of F_1 and of the mixture $\frac{1}{2}F_0 + \frac{1}{2}F_1$ from the upper α tail mass.
- (c) Compute VaR_α of F_0 and F_1 and explain why the example does not contradict the joint elicibility of (VaR, ES) .
- **Report** the three ES values; **answer:** what does this imply for ranking ES forecasts alone?
- Model: A6

A10: Null Moments of the Cumulative Violation [Proposed]

Solution discussed in class

Task

- ▣ **Context:** under a correct forecast distribution the PITs $u_t = F_t(L_t)$ are i.i.d. $U(0, 1)$; the cumulative violation of [Du & Escanciano \(2017\)](#) is $H_t = \frac{1}{\alpha}(u_t - (1 - \alpha))1\{u_t > 1 - \alpha\}$, with $\alpha = 2.5\%$.
- ▣ (a) Compute $E[H_t]$ and $E[H_t^2]$ by integrating over $u \in (1 - \alpha, 1)$, and deduce $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$.
- ▣ (b) Check both moments by simulating $4 \cdot 10^6$ uniform draws.
- ▣ (c) Compute the standard error of $\bar{H}$ for $T = 250$ and the value above which $\bar{H}$ rejects H_0 at 5% (one-sided, Normal approximation).
- ▣ **Report** the moments, the simulation check and the critical value; **answer:** why does H_t carry more information than I_t ?
- ▣ Model: A7

B1: Backtesting Six VaR Models on the S&P 500 (1/2) [Solved]

Task

- **Context:** S&P 500 daily closes (`data/market/GSPC.INDX.csv`), losses $L_t = -\ln(P_t/P_{t-1})$; one-day forecasts of the six models of Lecture 8 (HS, Normal, Student-t, GARCH-t, FHS, GARCH-EVT) on a rolling 1000-day window, parameters re-estimated every 20 days; evaluation from January 2007 to 18.09.2026, $T = 4\,959$ days.
- (a) For each model count the breaches of VaR 1% and compute the Kupiec p -value, both asymptotic (χ^2_1) and exact (two-sided Binomial).
- (b) Compute the Christoffersen independence and conditional coverage p -values [Christoffersen \(1998\)](#).
- (c) Fit the Weibull duration model [Christoffersen & Pelletier \(2004\)](#) with censored first and last durations; report the shape b and the LR p -value for $H_0: b = 1$.

B1: Backtesting Six VaR Models on the S&P 500 (2/2) [Solved]

Task

- (d) Run the DQ test [Engle & Manganelli \(2004\)](#): regress Hit_t on a constant, four lagged hits and VaR_t and compare DQ with χ^2_6 .
- (e) Estimate the size of the battery: simulate 1000 paths from a GARCH(1,1)-t fitted to the S&P 500 (1990–2026), backtest the true VaR 1% at $T = 250$ and $T = 5000$, and record the rejection rates at 5%.
- **Report** one table (breaches, rate, five p -values per model) and the simulated sizes.
- **Answer:**
 - ▶ Which models pass all tests, and which test is decisive?
 - ▶ Why can a model with the right number of breaches still fail?
 - ▶ Given the simulated size, how would you read a DQ p -value of 0.03 at $T = 250$?

B1: Backtesting Six VaR Models on the S&P 500: Solution (1/2) [Solved]

Model	x	Rate (%)	p_{uc}	p_{exact}	p_{ind}	p_{cc}	b / p_{dur}	p_{DQ}
HS	83	1.67	<0.001	<0.001	<0.001	<0.001	0.52 / <0.001	<0.001
Normal	134	2.70	<0.001	<0.001	<0.001	<0.001	0.56 / <0.001	<0.001
Student-t	99	2.00	<0.001	<0.001	<0.001	<0.001	0.53 / <0.001	<0.001
GARCH-t	88	1.77	<0.001	<0.001	0.095	<0.001	0.94 / 0.479	<0.001
FHS	61	1.23	0.116	0.127	0.046	0.040	0.93 / 0.514	<0.001
GARCH-EVT	56	1.13	0.370	0.395	0.028	0.060	0.88 / 0.240	<0.001

- Procedure: build the hit series I_t for each model, then apply the five tests to the same series; all p -values are at the 5% level.
- (a)–(c): only GARCH-EVT passes Kupiec, conditional coverage and durations; FHS fails p_{cc} narrowly; HS, Normal and Student-t fail the duration test with $b \approx 0.5$ (clustered breaches).
- With 4959 days asymptotic and exact Kupiec p -values agree; in 2008 alone ($T = 253$, $x = 8$) they are 0.0058 and 0.0086.

B1: Backtesting Six VaR Models on the S&P 500: Solution (2/2) [Solved]

Solution

- (d) DQ rejects all six models, including GARCH-EVT ($p < 0.001$): hits are predictable from lagged hits and from the VaR level, a conditional failure invisible to the counts.
- (e) Simulated size at 5%, $T = 250$: POF 9.8%, CC 1.2%, durations 14.8%, DQ 8.6%; $T = 5000$: POF 4.9%, CC 3.4%, durations 4.6%, DQ 8.8%.
- Reading: in one year the tests are mis-sized (POF too often, CC too rarely); DQ stays oversized even at 5000 days, so a DQ p -value of 0.03 at $T = 250$ is weak evidence and needs a simulated critical value.
- Economic conclusion: a model with the right number of breaches (FHS) can still fail on timing; only filtered models with a flexible tail survive most tests.

B2: Traffic Light for GARCH-t on the BET [Proposed]

Solution discussed in class

Task

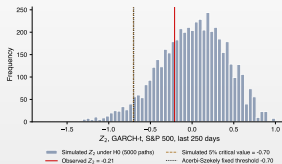
- ▣ **Context:** BET index (data/market/BET.csv), VaR 1% and VaR 2.5% of GARCH-t (rolling 1000-day window, refit every 20 days), 2007 to 18.09.2026; Basel zones from A5.
- ▣ (a) Count the VaR 1% exceptions in the last 250 days; give the zone, the cumulative Binomial probability $P(X \leq x)$ and the FRTB multiplier.
- ▣ (b) Count exceptions in every rolling 250-day window since 2007; report the share of days in each zone and the worst window with its end date.
- ▣ (c) Count the VaR 2.5% exceptions in the last 250 days and compare with the FRTB desk limit of 30.
- ▣ **Answer:** would a supervisor have penalised this model since 2007? Is time in the yellow zone evidence against the model or bad luck (use A5)? What does the traffic light ignore that Z_2 would capture?
- ▣ Model: A5 (zones), B1 (breach series)

B3: ES Backtests with Bootstrap and Simulation [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), forecasts of (VaR 2.5%, ES 2.5%) of GARCH-t, FHS and HS, 2007 to 18.09.2026, $T = 4\,959$ days (setting of B1).
- (a) On breach days compute the exceedance residuals $e_t = (L_t - \text{ES}_t)/\sigma_t$ [McNeil & Frey \(2000\)](#); report their number, mean and one-sided t -statistic, and compare the bootstrap p -value (5000 resamples of the centred residuals) with the Student- t p -value.
- (b) Compute Z_2 [Acerbi & Szekely \(2014\)](#) on the full sample and on the last 250 days; obtain p -values and 5% critical values from 5000 paths simulated from each model's forecast distributions.
- **Answer:** why is the bootstrap preferred for the residuals? Does the simulated 5% critical value for 250 days agree with the fixed -0.70 ? Why is the full-sample critical value much closer to zero?

B3: ES Backtests with Bootstrap and Simulation: Solution (1/2) [Solved]



Model	n	$\bar{e} / t$	p_{boot} / p_t	$Z_2 (T = 4959)$	$p / \text{crit.}$	$Z_2 (250 \text{ d}), p$
GARCH-t	192	0.106 / 1.80	0.023 / 0.037	-0.618	<0.001 / -0.153	-0.210, 0.289
FHS	130	0.093 / 1.21	0.096 / 0.114	-0.091	0.154 / -0.150	0.142, 0.602
HS	156	0.363 / 2.88	<0.001 / 0.002	-0.405	<0.001 / -0.154	0.723, 0.975

- Procedure: (a) centre the residuals, resample them 5000 times and read the share of bootstrap means above the observed mean; (b) simulate $L_t^* = F_t^{-1}(U_t)$ day by day and recompute Z_2^* .

B3: ES Backtests with Bootstrap and Simulation: Solution (2/2) [Solved]

Solution

- (a) GARCH-t: $\bar{e} = 0.106$, $t = 1.80$, bootstrap $p = 0.023$ versus Student- t $p = 0.037$: the residuals are skewed, so the t approximation is poor and the bootstrap is preferred.
- (b) Full sample: GARCH-t ($Z_2 = -0.618$, $p < 0.001$) and HS are rejected at 5%; FHS passes ($p = 0.154$).
- Last 250 days: the simulated 5% critical value for GARCH-t is -0.705 , close to the fixed -0.70 ; with only 6 breaches no model is rejected.
- Reading: the null distribution of Z_2 shrinks like $1/\sqrt{T\alpha}$, so the full-sample critical value (-0.153) is much closer to zero; economically, the thin t tail underestimates the average tail loss.

B4: Z_2 by Simulation on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ **Context:** BET (BET.csv, from 2007) and Bitcoin (BTC-USD.CC.csv, from 2017) to 18.09.2026; (VaR 2.5%, ES 2.5%) forecasts of GARCH-t and GARCH-EVT (rolling 1000-day window).
- ▣ (a) Compute Z_2 on the full evaluation sample, with its p -value and 5% critical value from 5000 paths simulated from the forecast distributions.
- ▣ (b) Repeat on the last 250 days, report the number of breach days, and compare the critical value with the fixed -0.70 .
- ▣ **Answer:** which model would you trust for ES on the BET and on Bitcoin? Why does the verdict change between the full sample and the last year?
- ▣ Model: B3

B5: Diebold–Mariano with HAC Variance [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), FZ0 losses [Patton, Ziegel & Chen \(2019\)](#) of the (VaR 2.5%, ES 2.5%) forecasts of the six models, 2007 to 18.09.2026 (setting of B1).
- (a) Compute the average FZ0 loss of each model and identify the best model.
- (b) Test GARCH-t, HS and GARCH-EVT against FHS with the DM statistic [Diebold & Mariano \(1995\)](#) and a Newey–West variance [Newey & West \(1987\)](#) with 9 lags; give the two-sided p -values.
- (c) Compute the naive t -statistic that ignores autocorrelation and the ratio of the long-run variance to the ordinary variance.
- (d) Run the Giacomini–White conditional test [Giacomini & White \(2006\)](#) with $h_{t-1} = (1, \Delta L_{t-1})$ and χ^2_2 critical values.
- **Answer:** which differences are significant? When does ignoring autocorrelation overstate significance? Does conditioning change the verdict for GARCH-EVT versus FHS?

B5: Diebold–Mariano with HAC Variance: Solution [Solved]

Solution

- (a) Average FZ0: FHS -3.551 , GARCH-EVT -3.548 , GARCH-t -3.512 , HS -3.134 , Student-t -3.036 , Normal -2.958 ; FHS is best.
- (b) $DM = \bar{d} / \sqrt{\widehat{LRV} / T}$. GARCH-t minus FHS: $\bar{d} = 0.0388$, $DM = 2.65$, $p = 0.008$: GARCH-t is significantly worse at 5%.
- HS minus FHS: $DM = 4.70$ ($p < 0.001$), significantly worse; GARCH-EVT minus FHS: $DM = 1.24$ ($p = 0.216$), no significant difference.
- (c) Naive t : GARCH-t 2.72, HS 7.18; for HS the long-run variance is 2.34 times the ordinary one, so the naive statistic is inflated by about $\sqrt{2.34}$: HS losses cluster in crises.
- (d) GW: GARCH-EVT 7.38 ($p = 0.025$), GARCH-t 7.59 ($p = 0.022$), HS 53.38 ($p < 0.001$); the DM tie of GARCH-EVT becomes a conditional rejection: yesterday's loss differential predicts which method wins today.
- Economic conclusion: FHS and GARCH-EVT are equally good on average, but their relative performance varies over time; the FZ0 gaps of GARCH-t and HS are statistically robust.

B6: Comparing Models on Bitcoin and EUR/RON [Proposed]

Solution discussed in class

Task

- ▣ **Context:** Bitcoin (BTC-USD.CC.csv, from 2017) and EUR/RON (BNR reference rate, from 2009) to 18.09.2026; FZ0 losses of the six models (setting of B5).
- ▣ (a) Find the best model in each market and test every other model against it with DM tests and Newey–West variance.
- ▣ (b) Compute the MCS p -values [Hansen, Lunde & Nason \(2011\)](#) with the T_{max} statistic, a block bootstrap (blocks of 10 days) and 1000 replications; list the 90% MCS.
- ▣ (c) On Bitcoin, VaR 2.5%, compute the mean elementary quantile scores [Ehm, Gneiting, Jordan & Krüger \(2016\)](#) of HS and GARCH-EVT on a grid of thresholds θ and check whether the Murphy curves cross.
- ▣ **Answer:** why is the MCS so large for Bitcoin and so small for EUR/RON? Relate it to the effective tail sample αT [Pele & Mazurencu-Marinescu-Pele \(2026\)](#).
- ▣ Model: B5

B7: Conformal VaR in Calm and Crisis Periods [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), 2007 to 18.09.2026; score $s_t = L_t / \hat{\sigma}_t$ with $\hat{\sigma}_t$ the standard deviation of the previous 250 returns; calibration on the last 250 scores; crisis days: previous 20-day volatility in its top 10% (known in advance).
- (a) Build the split conformal VaR 5% and VaR 1% and compute the breach rates overall, on calm days and on crisis days, with the Kupiec p -value.
- (b) Repeat on simulated i.i.d. Student- t_4 data of the same length (exchangeable by construction).
- (c) Apply the Christoffersen independence test to both breach series.
- **Answer:** is the marginal guarantee met on real data? What changes between real and exchangeable data: coverage or independence?

B7: Conformal VaR in Calm and Crisis Periods: Solution [Solved]

	All	Calm	Crisis	p_{uc}	p_{ind}	i.i.d.: all / calm / crisis
VaR 5%	5.34	5.15	7.06	0.272	<0.001	4.74 / 4.73 / 4.84
VaR 1%	1.29	1.21	2.02	0.049	0.001	0.73 / 0.76 / 0.40

- Procedure: for each day take the k -th order statistic of the last 250 scores, $k = \lceil 251(1 - \alpha) \rceil$, multiply by $\hat{\sigma}_t$, and record the hit; rates in %; crisis = 496 days.
- (a) VaR 5%: 5.34% overall (Kupiec $p = 0.272$, not rejected), but 7.06% on crisis days.
- (b)–(c) i.i.d. data: calm 4.73%, crisis 4.84%, independence holds ($p = 0.965$); real data: independence rejected ($p < 0.001$).
- Reading: the marginal guarantee is met; volatility clustering breaks exchangeability, so breaches cluster and crisis coverage fails.

B8: Conformal VaR on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- **Context:** BET (BET.csv) and Bitcoin (BTC-USD.CC.csv) to 18.09.2026; VaR 5% by split conformal, ACI with $\gamma = 0.005$, and GARCH-t; calm and crisis days defined as in B7.
- (a) Compute the breach rates overall, on calm days and on crisis days for the three methods in both markets.
- (b) Identify, for each market, the method closest to 5% on crisis days.
- **Answer:** does ACI fix conditional coverage or only the long-run average? Would you use a conformal VaR for regulatory capital?
- Model: B7

B9: Parametric Bootstrap Under Estimation Risk [Proposed]

Solution discussed in class

Task

- **Context:** data-generating process GARCH(1,1)-t with the parameters fitted to the S&P 500 (GSPC.INDX.csv, 1990–2026), as in B1(e); estimation window $R = 1000$ days, fixed scheme (estimate once, then forecast).
- (a) Simulate 400 paths of length $R + 5000$; on each, re-estimate GARCH-t on the first R days, forecast VaR 1% for the next P days, and run the Kupiec test for $P = 250, 1000$ and 5000.
- (b) Compute the rejection rate at 5% with the estimated parameters and, on the same paths, with the true parameters.
- (c) Compute the standard deviation of the breach rate across paths and compare it with the Binomial value $\sqrt{\alpha(1 - \alpha)/P}$.
- **Answer:** how does the distortion grow with P/R (relate to [Escanciano & Olmo \(2010\)](#))? Which critical values would you use for $P/R \approx 5$?
- Model: B1 (part e)

C1: Which Risk Model Would You Certify for a Romanian Bank's Trading Book? (1/2) [Proposed]

Solution discussed in class

Task

- ▣ **Context:** a trading book in lei, half in the BET index (data/market/BET.csv) and half in euro valued at the BNR EUR/RON reference rate; daily data from 2005, evaluation from 09.07.2009 to 18.09.2026; the six models of Lecture 8 with a rolling 1000-day window.
- ▣ You are the model validator: you must certify one VaR/ES model for FRTB reporting, combining regulatory checks with statistical evidence.
- ▣ (a) Join the two price series on common days, build the portfolio value and its daily log returns.
- ▣ (b) Forecast VaR 1%, VaR 2.5% and ES 2.5% with the six models and backtest them with at least three of: Kupiec, Christoffersen, durations, DQ, Z_2 .
- ▣ (c) Place each model in the traffic light for its worst and for its last 250-day window, and check the FRTB desk limits.

C1: Which Risk Model Would You Certify for a Romanian Bank's Trading Book? (2/2) [Proposed]

Solution discussed in class

Task

- ▣ (d) Rank the models with the FZ0 loss, DM tests with Newey–West variance and the 90% MCS.
- ▣ (e) Report the size and power of your test battery at $T = 250$, as in B1(e).
- ▣ **Deliverable:** a Quantlet folder in the team repository and one slide with the recommended model, the evidence table and the main limitation.
- ▣ **Answer:** would your choice survive a year like 2008 or 2020?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily S&P 500 closes in a pandas Series P. Write Python for a rolling 250-day historical-simulation VaR 1%, count the exceptions and run the Kupiec and Christoffersen tests."
- "My last 250 days have 7 exceptions. What does that mean for the model?"

The AI answer: text

- Losses $L_t = -r_t$; VaR_t is minus the 1% quantile of the last 250 returns; exception: $L_t > \text{VaR}_t$
- `lr_ind(I)` returns the Christoffersen independence statistic LR_{ind} Christoffersen (1998)
- Your last 250 days: $x = 7$, $\text{LR}_{uc} = 5.50$, $p = 0.019$: correct coverage is rejected at 5%
- Basel traffic light: 7 exceptions is the red zone, so the model must be withdrawn BCBS (1996)

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np, scipy.stats as stats
r = np.log(P).diff().dropna().values
W, p = 250, 0.01
VaR = np.array([-np.quantile(r[t-W:t+1], p)
                 for t in range(W, len(r))])
I = (-r[W:] > VaR).astype(int)
T, x = len(I), I.sum()
ph = x / T
LR_uc = -2*((T-x)*np.log(1-p) + x*np.log(p)
           - (T-x)*np.log(1-ph) - x*np.log(ph))
LR_cc = LR_uc + lr_ind(I)
p_uc = stats.chi2.sf(LR_uc, df=1)
p_cc = stats.chi2.sf(LR_cc, df=1)
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ Which returns are known at the close of day $t - 1$, when VaR_t is reported?
- ▣ How many restrictions does the conditional coverage test impose?
- ▣ Compare the zone with A5 and with the Basel table [BCBS \(1996\)](#)

Key Takeaways

- ▣ Kupiec counts, Christoffersen, durations and DQ look at timing: always use both; know their size and power at your T
- ▣ Estimated parameters with P/R large distort χ^2 critical values: bootstrap with re-estimation
- ▣ The traffic light is a Binomial test; it can let an inaccurate model through one time in eight
- ▣ ES tests need the forecast distribution: bootstrap and simulation are the honest way to get p -values
- ▣ Rank with FZ0 + DM (HAC) + MCS; significance needs many tail days
- ▣ Conformal coverage is marginal; in crises it fails like an unconditional model

Take-Home Assignment (Team Project)

- Choose a portfolio not covered in class (a BVB stock, a currency or a crypto-asset); forecast VaR 1% and ES 2.5% with at least three models
- Backtest them with Kupiec, Christoffersen, durations and Z_2 ; place each model in the traffic light for its worst 250-day window
- Rank them with FZ0, Diebold–Mariano tests (HAC) and the MCS
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

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